


Keywords

Following tea table / Adjustable surface / Portable travel box

Tools

Figma / Blender / STM32H743 / OV2640

**Introduction**

NEARBY is a following, height-adjustable tea table for young adults' downtime. Tea, a book and a phone stay together between indoor seats. For a short stay, the surface lowers and locks into a compact box with a pull handle and side carry grips.

**Goal**

FOLLOW starts an indoor session; STAY parks the table and HEIGHT adjusts the surface. PACK MODE guides clearing, lowering and locking before manual transport. At the destination, unpack and review the new room before following.

Through this project,

I explore how responsive furniture can let a tea break move naturally with a person's changing place and posture.

Original lift-box prototype · Portable transport proposal

Modelled packing envelope · Vehicle fit testing next

# Background

## What is a followable tea table?

A following, height-adjustable tea table keeps leisure items nearby at home. For a short stay, it lowers into a locked, manually towable travel box.

Relax on the balcony or by the sofa without carrying the table. When leaving, store loose items, lock the lowered surface and drawers, then tow or lift the box into the car.

## Project Context: Home + Short-Stay Living







**HOME**
Living room + enclosed balcony

**STAY**
Short-stay accommodation

**BREAK**
Tea · reading · relaxation

**FOLLOW + LIFT**
Move with you · fit your seat

**Young Adults**
Effortless downtime, wherever you sit

## Main Characteristics

**Move the break with you**
Keep a cup, book and phone together as you switch between your indoor resting places.

**Fit different ways of sitting**
Use a lower surface beside the sofa or raise it for a chair. Adjust the height after parking.

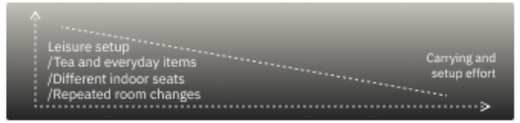
**Pack for the next stay**
Lower and lock the table for travel. A pull handle and side grips support manual towing and two-hand loading.

## Current Situation

A relaxing break can involve small but repeated tasks: carrying drinks, moving a table and resetting a place to sit. The project explores how furniture could move with the user.

| Carrying | Moving | Resetting |
|---|---|---|
| / cups + phone + book<br>/ occupied hands<br>/ repeated trips | / table left behind<br>/ changing rooms<br>/ clearing a route | / new sitting position<br>/ mismatched height<br>/ interrupted leisure |

## Opportunity Direction




**Fixed furniture**

The person carries items and moves the table.

**Following furniture**

The table follows, parks and adjusts to the seat.

Design aim: bring a complete break along, between seats and between stays.
Combine controllable following and parked-height adjustment with compact, manually handled transport.




Basis: creator's brief - 2026-10-10. Photos: Pexels P2, P4. Cover: AI concept simulation. Research assumptions; benefits not measured.

# Scenario Research

Home and short-stay leisure · Scenario study from the brief
Questions for future observation, not completed fieldwork.

## Pain Point Themes

### Human–Object Interaction: Carrying a complete tea break



Several items to carry

Hands already occupied



Table stays in one room

Extra trips for essentials



Drink needs a stable surface

Balancing a tray

TEA BREAK
RESET

Different seat heights

Setup starts over again

Illustrative context photographs · carrying, changing places and settling in are design hypotheses.

### Human–Space Interaction: Switching the place to relax



1. Indoor movement



2. Doorway transitions



3. Leisure setup

Relax across rooms.

Pack, manually tow and load for the next stay.

New study questions.

### Human–Furniture Interaction: What should change?




1. Move with me




2. Fit my seat




3. Let me pause

*Could following and height adjustment reduce carrying and setup?*

*Compare with a manual table.*

Basis: creator's leisure brief. Context images: Pexels P1–P8; illustrations, not participant evidence.

# User Research & Insights

## User Groups

**Home lounger**
Young adult · tea, reading, relaxing
Switches between favourite seats

**Short-stay guest**
Young adult · short-stay travel
Wants a towable, stowable setup

**Shared-home user**
Young adult · shared living spaces
Wants clear, controllable motion

Recruitment targets, not completed interviews.

## Problem Redefinition

What interrupts an easy indoor break?
Three questions derived from the brief.

**Carrying the whole setup**
Can cup, book and phone move together without repeated trips?

**Changing resting places**
Can the table follow when comfort or light sends me elsewhere?

**Fitting the next seat**
Can a parked surface adapt to a sofa, chair or relaxed posture?

**Design opportunity: less carrying indoors + compact transport between stays.**

## Design Lens: Leisure Interruptions

- Observe tea setup, room changes, packing and car loading.
- Compare a manual side table and luggage; measure effort, stability and handling.

POSSIBLE EXPERIENCES

Ready · Carrying · Interrupted · Relaxed

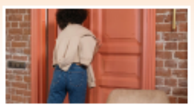
Setup tasks → Perceived effort → Break interrupted

Explore when a change of place interrupts leisure and what movement people would accept.
The relationships and feelings shown are hypotheses for a future study.

Basis: brief-led scenario analysis · validate through observation and comparative trials.

## User Journey & Effort Map

BASELINE SCENARIO / Balcony tea → living-room rest with a fixed table · hypothesis

| | 1 PREPARE | 2 CARRY | 3 SETTLE | 4 RELOCATE | 5 RETURN | 6 RESET | 7 ADJUST | 8 RELAX |
|---|---|---|---|---|---|---|---|---|
| Possible questions | *What do I need for a break?* | *Can I carry everything?* | *Where shall I sit first?* | *Shall I move to the sofa?* | *Did I leave my phone?* | *How do I move the table too?* | *Is this height comfortable?* | *Can I relax again now?* |
| Practical demands | Gather the tea setup | Balance several items | Choose a resting place | Change the setting | Retrieve a forgotten item | Carry and reposition | Fit the new seat | Resume the break |
| Emotional experience | 😐 | ☹ | 😐 | 🙂 | 😐 | ☹ | 😐 | 🙂 |
| Actions |  Tea preparation context |  Tray + small essentials |  Balcony break context |  Room-to-room movement |  Doorway / return route |  Next seated tea setup |  Seated reach context |  Sofa reading context |

# Design Concept

## Solution Integration

**Your tea break, on the move.**
Following + adjustable furniture for downtime at home and on a short stay.

**Young adult / direct controls**
FOLLOW · STAY · HEIGHT

**Personal app**
Save places · Set height · Check charge

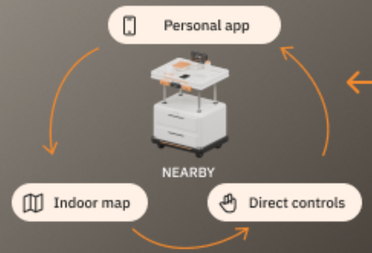
**Tea table + indoor map**
Tea setup · Parking places · Visible state

## Design Goals

01 **Bring the whole break along.**
Keep a cup, tea flask, book and phone on one mobile surface.

02 **Move between indoor leisure spots.**
Start FOLLOW, change rooms and stop beside the next seat.

03 **Fit the seat. Pack for the next stay.**
Adjust after parking; lower and lock for manual transport.




### System Integration

The table and app share following, height and packing states.
Stop before packing, then use the pull handle and side grips.

### Technology

Prototype: wheeled cabinet + lift.
Proposed: locks, grips + towing.

# Technical Research

## Embedded Vision + Motion Control

**On-device vision [1]**
Local face matching links the device to a selected user. Identity matching and continuous person tracking are separate engineering tasks.

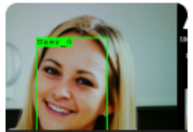
**Four-point electric lift [2]**
Four A2 motor-driven electric pushrods raise and lower the tabletop. Coordinated travel, stopping and load stability need verification.

**Following + phone commands [3, 4]**
The creator reports a powered base that follows a person. The proposed phone UI adds start, pause and parked-height controls.

## Basic Structure / Functional Architecture




## Technology Applications / Reference Examples



**01 Recognise**
On-device face matching
ST application [1]



**02 Lift**
Electric linear motion
Actuator reference [2]

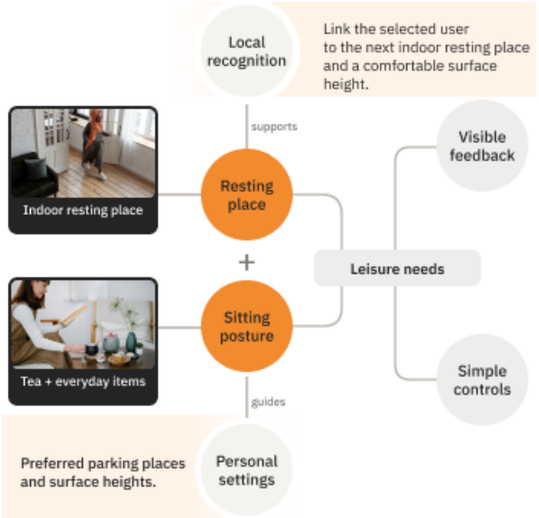


**03 Move**
Motor-driven wheelbase
Drive reference [3]

[1] ST / FP-AI-FACEREC1 [2] Bansbach / Electric actuators
[3] Pololu / 3pi+ 2040 [4] LINAK / Desk Control App

# Concept Development

## How Does NEARBY Help?




## Advantages to Validate

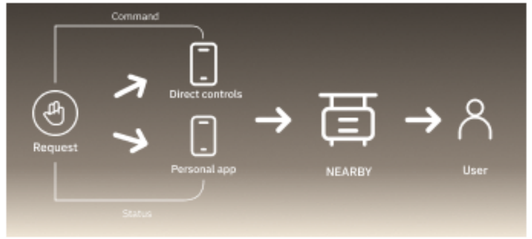
| Continuity | Control | Less carrying | Comfort | Simple setup |
|---|---|---|---|---|
| Bring the break with you | Start and pause at any time | Keep essentials together | Fit the next sitting position | Save a place and height |

## Core Topic

A tea break that moves with you.
Follow indoors, park and adjust;
pack for manual travel between stays.

**REQUEST**
- FOLLOW begins a session. STAY stops. HEIGHT adjusts while parked.

**FEEDBACK**
- Following / Paused / Ready
- Visible feedback for the next action.




1 Choose FOLLOW, STAY or HEIGHT.
2 Park before reaching or changing height.
**3 Stop before packing; clear the tray, lower and lock. Tow manually with following disabled.**

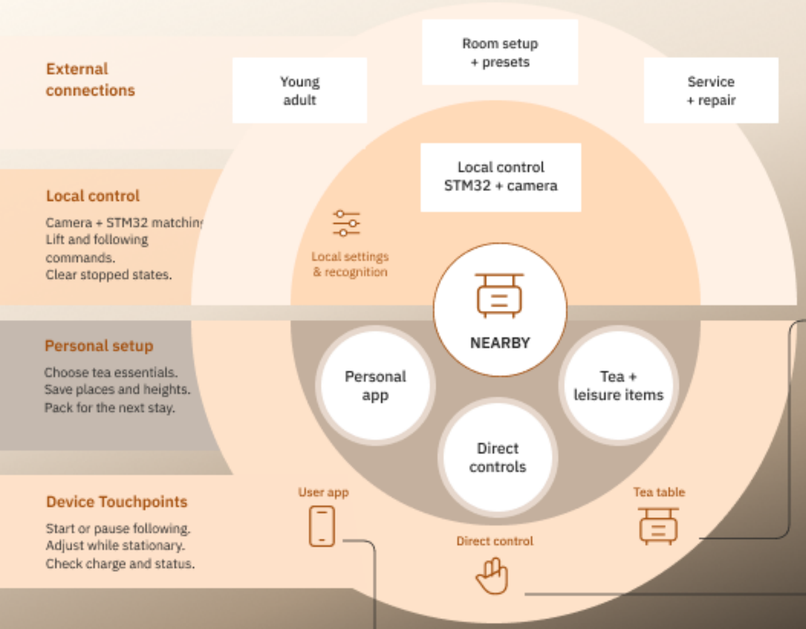
## Essential Information

**Next-action information**
Show only what matters at this step.

| | | |
|---|---|---|
| Cup · tea flask<br>Book · phone | Indoor places<br>Clear routes | Following · parked<br>Height + battery |

# System Design

## System Map




# Touchpoint Definition

## Personal Touchpoint: A Complete Tea Break

Cup, book and phone travel together.
Direct controls show whether the table is following, paused or ready to use.



**One tea setup**
Keep cup, book and phone on the tray.

**Follow on request**
Start FOLLOW; use STAY to stop at a seat.

**Fit the seat**
Adjust only after the table is parked.

FOLLOW · STAY · HEIGHT

## Product Functional Architecture

- Keep leisure items together.
- Follow and park between seats.
- Adjust while stationary.
- Lower, lock and manually transport.

- FOLLOW: start a following session.
- STAY: stop; HEIGHT: set the surface.

- Save places and preferred height.
- Guide packing and manual checks.

**Device Touchpoints**

**A Tea table**
Mobile storage · adjustable tray
Visible movement and height state

**B Direct controls**
FOLLOW / STAY / HEIGHT
One-touch movement and adjustment

**C Personal app**
Places, presets and packing checks
Review status before transport

**Essential Functions**
Mobile tea storage with direct following and height controls.

**Following + Lift**
Carry the break between seats, then fit the surface height.

**Extended Functions**
Personal places, height presets and a guided packing checklist.

# Touchpoints + Environment

## Space Management

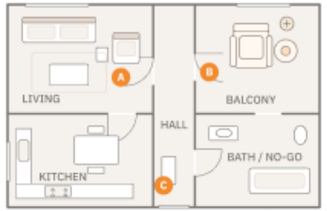
**Home + short-stay review**

- Identify living-room and balcony seats.
- Check doorway clearance and thresholds.
- Review the current accommodation.

**Adjustable settings**

- Save a few indoor parking places.
- Mark wet areas and door thresholds.
- Confirm a clear route for following.
- Pause if the target or route is unclear.

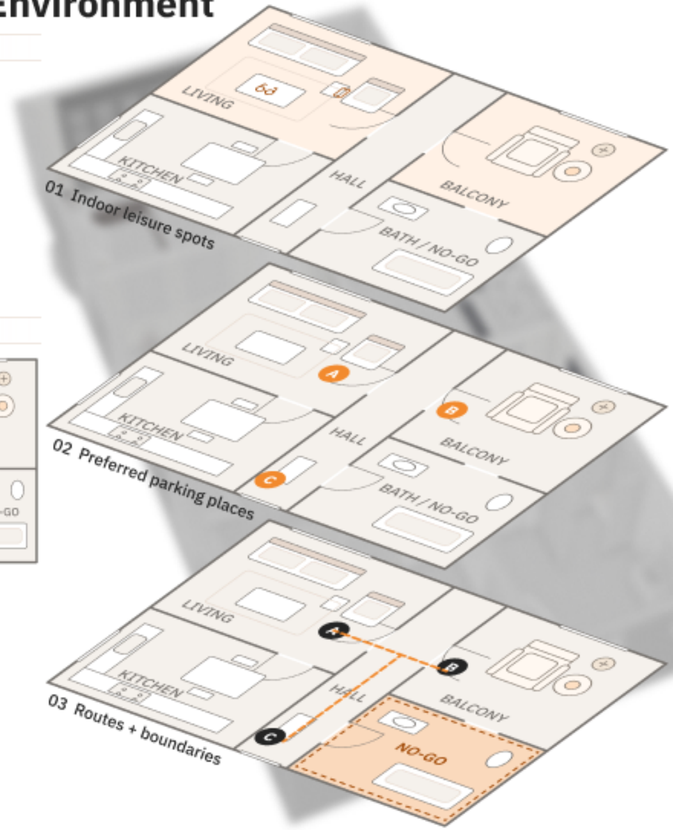
## Product Layout Principle




**A / Living-room sofa**
Park beside the sofa with tea nearby.
Adjust the surface for a relaxed seat and keep the walking route clear.

**B / Balcony chair**
Use an enclosed balcony on the same level.
Review the doorway threshold before enabling a following route.

**C / Charging place**
Return the table to a clear charging spot.
Review the layout when moving to a new short-stay accommodation.



# Device design

## Proposed component architecture



**Leisure mode → packed travel box**

New: pull handle, side grips and transport locks.
Vehicle fit and handling strength require tests.



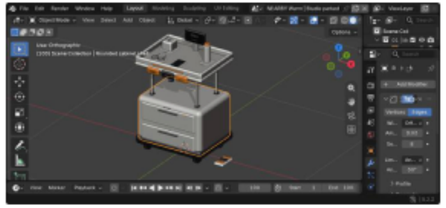
01 Lower + lock



02 Manual tow



03 Car / AI concept



Blender modelling view · concept renders




AI concept simulation · parked tea break; proposed controls and appearance

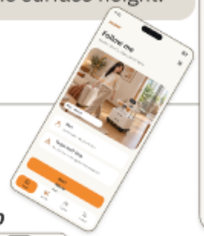
# App Design

## Personal leisure controls



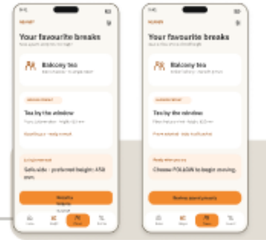
*Follow & adjust*

Start following.
Park by the next seat.
Set the surface height.



*Home*

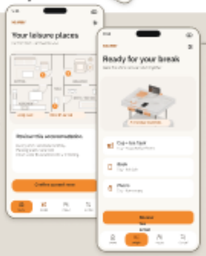


Save a tea-break preset
and recall its place
and preferred height.

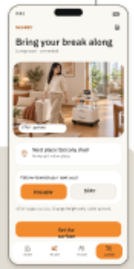
*Places & tea setup*

Review this room, save indoor parking places and keep tea, a book and phone ready to move.



*Direct control*

Start or pause, check charge and adjust a parked table.



*Pack Mode / Manual Transport*



**LEISURE → TRAVEL**

Clear → lower → lock
Turn following off.
Confirm checks yourself.

**TOW & LOAD**

Extend the pull handle.
Carry with both side grips.
Secure inside the car.

Proposed mode
Locks and car fit need testing.

## Visual Guidelines

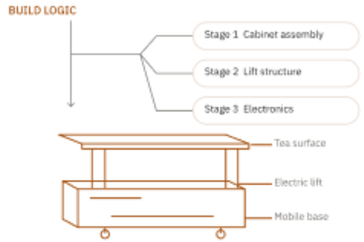
Font

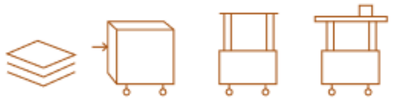
*IBM Plex Sans*

Colour

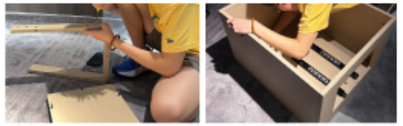
# Prototype | Following tea table

From a lift-box demonstrator to a leisure tea table: cabinet assembly, electric lift and control integration.



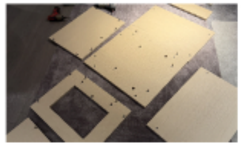
Panel kit — Cabinet → lift → integrated prototype



02–03 Assemble the cabinet and secure the drawer rails.



## *Mechanical assembly / steps 01–06*



01 Prepare the panels



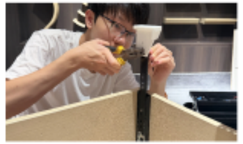
02 Fasten the cabinet



03 Fit the drawer rails



04 Mount lift hardware



05 Align the guide rails



06 Tighten the fasteners

## *Wiring preparation / steps 07–12*



07 Solder the control board



08 Prepare actuator connections



09 Wire the control board



10 Connect the cable set



11 Route cables inside the cabinet



12 Fit the display and connectio



## *Original lift-box demonstrator*

STM32H743 + OV2640 · fixed ROI
PC-tested core; board testing pending
Face matching supports pairing; tracking is separate.

# A break that follows




### *Reflection + Next Test*

Next, compare indoor carrying and travel setup with a manual table. Measure packed mass, grip loads, drawer retention, stop accuracy and drink stability. Check boot openings and lifting effort on a target car. The transport features are proposed; the original demo validates no vehicle fit.

AI concept simulation · movement behavior and drink stability not yet validated